

Section 01: Introduction & Requirements

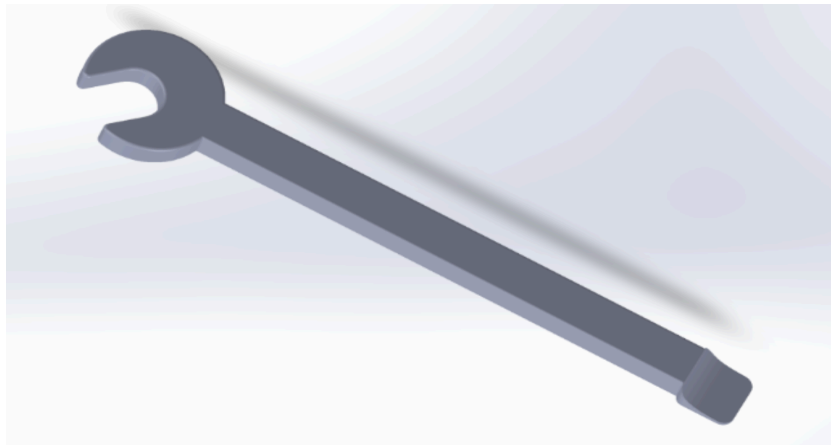
The following report documents the design, fabrication, and testing that Group 6 went through to produce a wrench design for Extra-Ultralight Biking LLC, a company who provides bikers with ultralight biking gear. Group 6 was informed by Extra-Ultralight Biking LLC that performance and weight are most important to them, and that they would like the weight limited to 1.15 oz if possible. Group 6 was also limited to 1.25" x 0.25" x 8" aluminum bar stock, with manufacturing operations limited to the basic capabilities of a milling machine. The handle was required to be 5", with 4" of a "grip". For some design iterations, part of the "grips" contained an angle, but the group decided that this could reasonably be considered a grip. The handle was also required to be able to withstand all required loads, while not being too difficult for the user to apply those loads. Finally, the handle was required to have a flat surface for the three point bend test, and have no sharp edges. The ultimate goal of the design was to produce a wrench which met these requirements while passing a 50N three point bend test, and a functional test with a 250 lb-in torque. To design a wrench capable of passing these tests, Group 6 used methods of by hand analysis using Euler-Bernoulli Beam Theory, and finite element analysis using Ansys. Group 6 aimed to have a factor of safety of greater than three for all analysis for the final design. Group six ultimately went through six iterations before settling on a final design.

Section 02: Concept Generation

Zack's Initial Concept:

Slight angle connecting the handle and wrench head to distribute stresses. Hex shape is not compliant with wrench requirements for 2 sides of contact. Fairly classic wrench design larger head with smaller handle size.

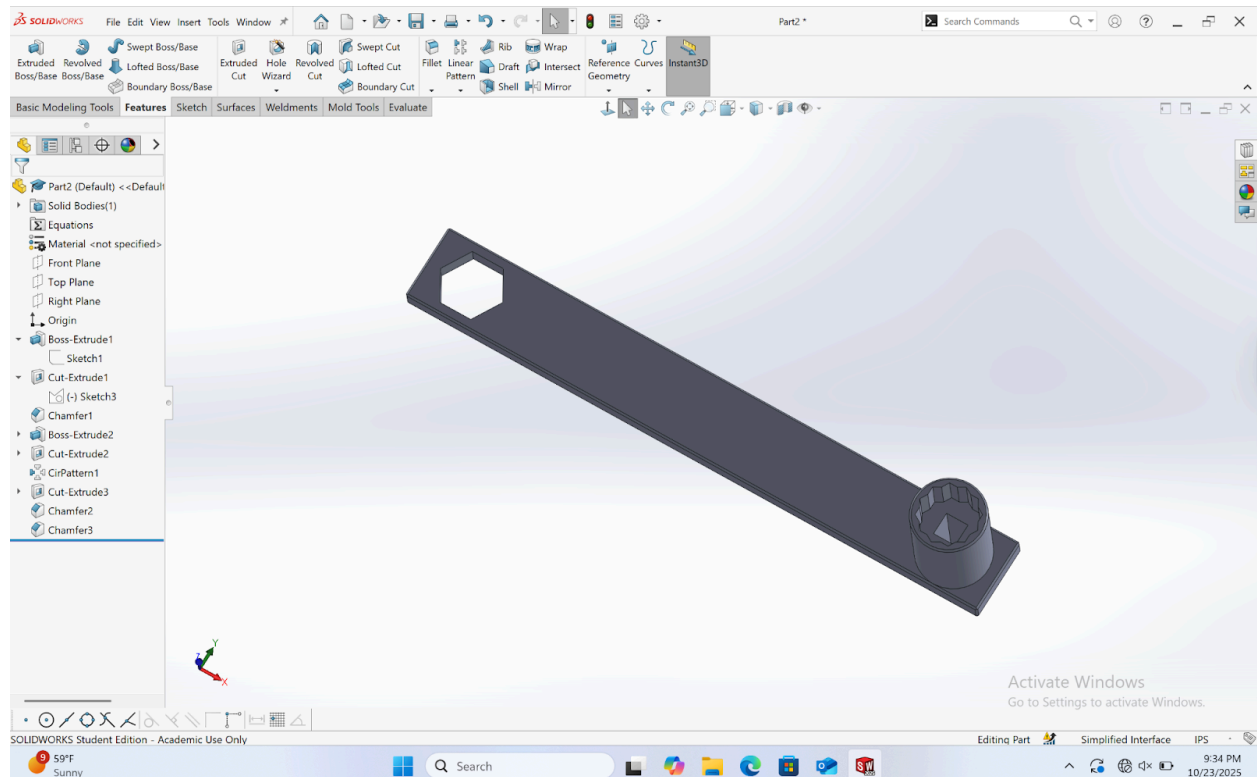
Ben's Initial Concept



This initial wrench design looks very typical. As in the final design, this design features a variant of the bike tire removal tool. The wrench met the size requirements however it was discarded for a number of reasons. First of all, the wrench was overweight. With weight being a driving factor,

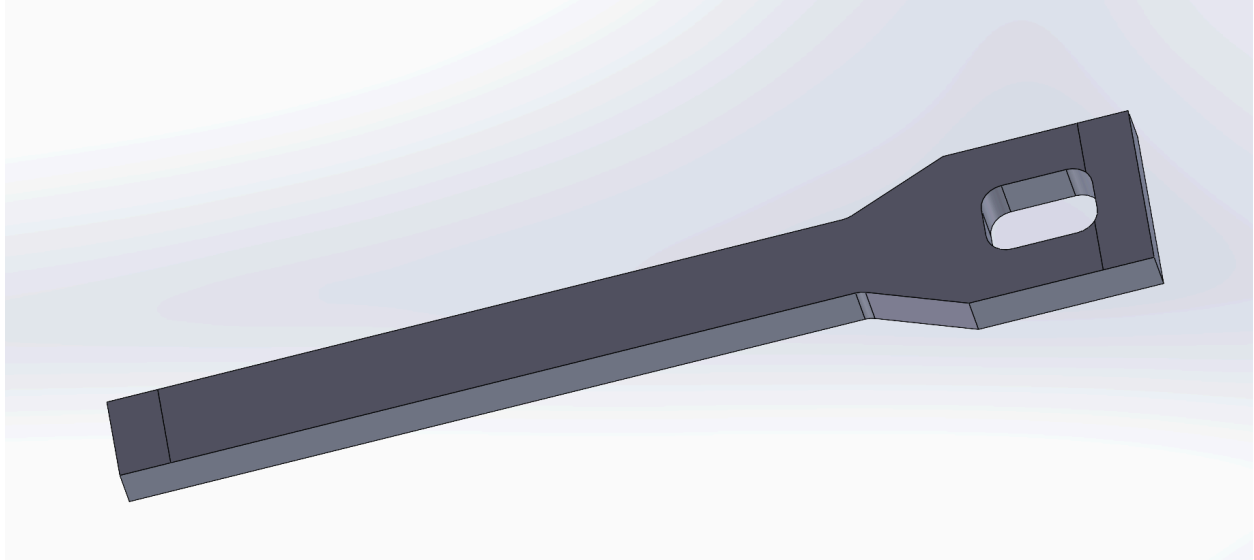
the design immediately fell short of what Group Six hoped for. Next, the handle was too long and narrow, thereby increasing the risk of yielding. Another problem with this design was that the rounded external profile on the head would be nearly impossible to machine using manual methods alone. Finally, the open head meant decreased structural support and introduced slippage potential.

Ari's Initial Concept:



Rectangular wrench which would be easy to machine (If socket wrench was 3D printed). This hex does not meet the requirements for two points of contact. Has a socket wrench as the secondary tool.

Isaac's Initial Concept:



Isaac's wrench resembles a typical wrench, but with much simpler geometries, and a closed head. The geometries were kept simple to allow for easy (and possible) machining. The head is closed for extra strength of the jaws. With slight modification to the width, this design could feasibly be under the goal weight. Group 6 ultimately chose this design for their first iteration

Section 03: By-hand Analysis Set-up Zack

The Following Assumptions were made for the by-hand analysis:

- Homogeneous Material, Plains remain planes
- Constant Cross-sectional Area
- Assume the Wrench can be modeled as a cantilever beam with an end load for the torque analysis.
- Assume the wrench can be modeled as a beam with simple supports and central load for the bending test.
- Bending test is Bending dominant and Torque test is shear dominant.

The objective of the by hand analysis is to develop initial dimensions for Group Six wrench design before the use of FEA to refine the design. The process began with finding the proper force to apply for the torque test:

$$T = F \times L \rightarrow F = T / L$$

Using the Free Body diagram, the resultant forces and moments were calculated. From these results Shear and Moment Diagrams were derived solving for the max Shear stress. Using that max shear stress the transverse shear stress formula was calculated to solve for the width and height of beam needed for torque test:

$$\tau = .57\sigma$$

$$3FS = \frac{\tau_{max}}{\tau_{allowed}}$$

$$\tau = \frac{VQ}{IB}$$

$$I = \frac{bh^3}{12} \quad Q = A\bar{y}$$

Next the bending stress was tested to solve for its respected width and height Using M_{Max} from the moment diagram:

$$\sigma_b = \frac{M_{max} y}{I}$$

$$I = \frac{bh^3}{12} \quad y = h$$

Using these values from the hand analysis initial dimensions of the wrench were formulated.

Overall, the by-hand bending and torque analyses give first-order predictions of the wrench's stress response in both bending and torsion. Despite simplifications, the models from the by hand analysis create a baseline for evaluating test results and assessing what dimensions allow the wrench to meet strength requirements.

Section 04: Finite Element Analysis Set-up Isaac

Group Six designed and implemented three finite element analysis simulations in Ansys static structural for each iteration of the part. All simulations used the material properties assumed in this project for Aluminum 6061, an elastic modulus of 68.9 GPa and a Poisson's ratio of .33. For all simulations, an element size of .001 m³ was used, and it appears that SOLID187 was the element type used for most simulations.. For the ideal torque loading case two fixed supports were used within the head of the wrench where the bolt head would come into contact with the wrench. This assumes that two opposite surfaces of the bolt head will remain fully in contact with the wrench head for the duration of loosening of the bolt. It also assumes that there will be no deformation at those surfaces (they are fixed). For loading, Group 6 added a force at a line along the width of the wrench at the end of the wrench handle which resulted in a torque about the bolt head of the required 250 lb-in. Modeling it this way is dissimilar to the loading conditions when a hand loosens a wrench, and would produce different bending moment and shear force distributions. That being said, modeling it this way results in an equal or higher bending moment along all all points on the wrench, and likely only a slightly smaller shear force. Based on this, Group 6 made the assumption that these boundary and loading conditions would be an appropriate way to model the wrench for this scenario. For the combined loading case Group 6 had the same boundary and loading conditions, with the addition of a 10 N load enacting a torque perpendicular to the existing torque. The 10 N force was located on a line along the height of the wrench, at the same location along the handle as the existing force. The assumption Group 6 made here is that 10 N at the end of the handle would be sufficient to model any incidental torques applied in this direction by the user of the wrench. The next finite element analysis simulation conducted by Group 6 was for the three point bend test. Group 6 chose to model this scenario by applying displacement supports to lines separated by five inches on the

face of the wrench which would be in contact with supports during the three point bend test. Displacement was set to zero in the X, Y, and Z directions. Although there would likely be small contraction during the three point bend test, Group 6 decided that this would likely be minimal enough as to not affect results if displacement supports were set to zero in all directions. A 50 N load was applied to the center of the wrench along a line. These loading and boundary conditions were created with the expectation and assumption that the applied force and supports would be in contact with the wrench in a small enough region that a line would be appropriate to model them with. For the solution section of Ansys, Von Mises equivalent stress was selected. All models are inherently prone to inaccuracies, and Group 6 chose simulation setups and assumptions to minimize them.

Section 05: Design Process

Group Six's Design process started with the initial design which was a very basic wrench design. This wrench had the basic necessities required by the project, a handle, and a $\frac{1}{4}$ -20 wrench socket. From the by hand analysis the group had a rough outline for the wrench from which the initial design was created. Once the initial design was finished Group six ran the Finite Element Analysis on the wrench in Ansys based on the by hand analysis, to test the effectiveness and durability of the wrench. This test came back showing large stress concentrators on the edge between the handle and the head of the wrench. Due to this stress concentrator, the maximum Von Mises equivalent stress was 190 MPa for the ideal torque case, and 197 MPa for the combined case. For the bending case, there was a Von Mises equivalent stress of 16 MPa. Ultimately it was the goal of Group 6 to get this above a factor of safety of three for all cases, which means an Equivalent Von-Mises stress of less than 90 MPa. To improve on this design it was figured that if the edge was blunted and the edge was found further down the wrench the stress would lower significantly. In other words the angle between the handle and the head was increased. Group Six thought this would help in part because it would increase the average height across the wrench, and decrease the bending moment at the stress concentrator. This added significant mass and to remain below the target weight the width was decreased from .23 in to .19. When simulated in Ansys, this iteration decreased the maximum Von Mises stress to 145 MPa for the ideal torque case and 155 MPa for the combined case. The stress increased slightly to 22.5 MPa for the bending case, likely due to the decreased width. While this reduced stress in that area Group 6 still believed that more improvements could be made. In an attempt to further reduce stress, Group six added two .5" wide, .09" deep pockets on each side. The reduced mass allowed us to increase the width from .19" to .21" in hopes of decreasing stress. The increase in width decreased the maximum Von Mises equivalent stress to 72 MPa in the ideal torque case, 81 MPa in the combined case, and 15 MPa in the bending case. The mass for this iteration was 31.98 grams. This iteration was very important because it showed Group 6 that removing material in the middle through slots or pockets, and adding this material elsewhere, was a practical way to reduce stress. In the fourth iteration, Group Six decided to move to a

design with a uniform height. This decision was made because it removed the stress concentrator at the angle between the head and the handle. This decision was also made after a discussion with the machine shop, where they learned that having an angle between the handle and head would be difficult and time consuming (although not impossible) to machine. For this iteration, the uniform height was .75", and the width was increased to .25". Group 6 made this possible by changing the two pockets present on each side in iteration three, to one large slot that ran across the handle. The simulation results for this iteration were 81 MPa for the ideal torque case, 100 MPa for the combined case, and 50 MPa for the bending case. This increase in stress meant that this iteration did not exactly accomplish what Group 6 thought it might have. That being said, they saw potential in this design if they kept building upon it. For their fifth iteration, Group 6 decided to try a unique design where they combined the slot along the handle and the head of the wrench, so that a bolt could be tightened or loosened anywhere along the length of the handle. To make this possible, Group 6 combined the slot and the head, and decreased the width of the slot to .44", the width of the bolt head. They also had to decrease the length from 6.8" to 6.2" to make up for the added weight to the decrease in slot width. These changes ultimately had very little impact on the simulation results, so Group 6 made the decision to scrap this design and move forward with a different one. For their final iteration, Group 6 decided to increase the height to 1" increase the slot width to .75", reduce the slot depth to .07", increase the total length to 6.55", and decrease the width from .25" to .21". The decrease in wrench width and increase in slot width were made so that Group 6 could increase the height of the wrench, which was very important to Group 6. Increasing the average height of the wrench was important to Group 6 because height is raised to the third power in the moment of area of a rectangular cross-section, and the moment of area in the denominator of the bending stress formula. This means that even small increases in height have the chance to decrease bending stress by a large amount. When brought to Ansys, these changes decreased the maximum Von Mises stress to 45 MPa in the ideal torque case, 73 MPa in the combined case, and 26 MPa in the bending case. This means that Group 6 had achieved their goal of having a factor of safety above three. Since this design also met the 32.6 gram weight requirement, they decided that this would be their final design for the wrench.

Section 06: Test Results - Ben

On the day of testing, the wrench passed the three point bending test without failure. This was expected as the team's finite element analysis for this loading case demonstrated a factor of safety greater than 10. At zero seconds the deformation value displayed was 0.6128mm. The max recorded deformation value was 1.31827mm. The difference between these values was

0.70547mm which represents the actual maximum deformation. This level of deformation occurred when the applied force reached 50N. The very last data point revealed a deformation value of 0.958mm however the corresponding applied force for that observation was roughly 2N not 0N. This means it is difficult to compute the total deformation with complete accuracy, especially given that the deformation values continue to decrease up until this cutoff point. Taking the difference between the final and initial values, a total estimated deformation of 0.3452mm is obtained. It is important to note though, that during the ramp-up a very similar value to 0.958mm was observed when 2N was initially reached, a difference of merely 0.1mm. In the context of structural performance, this value is likely negligible, indicating deformation within elastic recovery. That said, it is impossible to draw conclusions with absolute certainty.

The second test proved to be a major challenge. For this functional torque test, the bolt was first tightened in alignment with Group Six's specification criteria. Next it was time to test wrench performance. Although designed with a safety factor around 3.5, when the user applied torque to the wrench handle the bolt did not loosen. In fact, the wrench head stripped in multiple contact locations indicating visible plastic failure. One thing that was apparent during the test was that the wrench did not provide a perfectly snug fit around the bolt head. In addition, the dimensional thickness of the wrench was about double that of the bolt head.

Another unexpected failure was the measured weight of the final wrench during testing. In the testing facilities the wrench weighed in at 32.9 grams. This contradicted the value taken in the machine shop on the day of production which was 32.6 grams or 1.15 oz exactly pictured below.



Section 07: Discussion & Conclusion - Ari

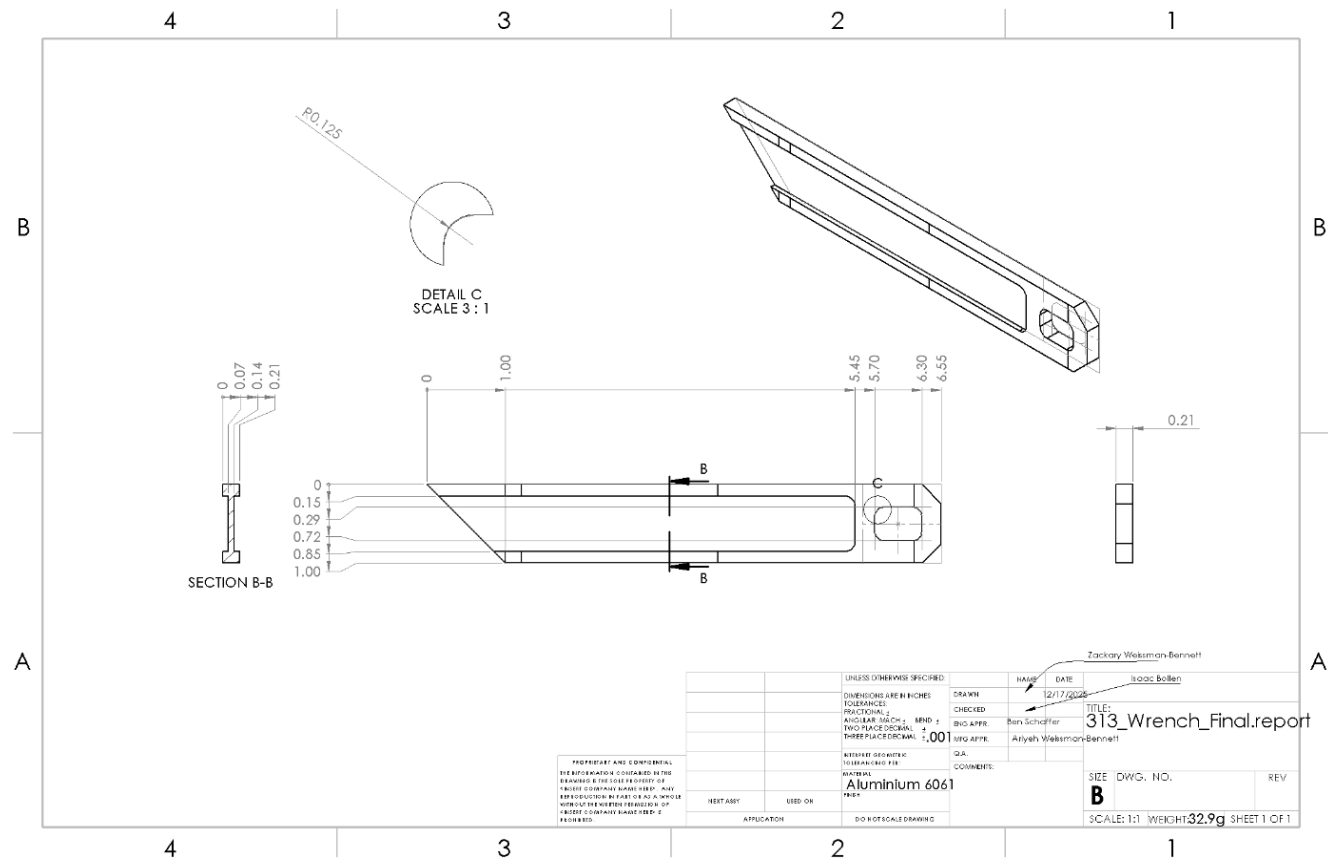
While Group Six did pass the bending test, the wrench ultimately stripped and failed during the Torque test. To start to understand why this may have happened when all of Ansys analysis showed large Factors of Safety begins with reimagining the Ansys setup. Group Six may have

been able to better predict such an outcome if the Ansys Static analysis had been done in a different way, perhaps by splitting the inside face of the wrench in half to more accurately simulate how the wrench would interact with the face. It is worth noting that during the Ansys analysis of the wrench a large stress concentration increased exponentially as the mesh increased. While Group six believes that this was a stress that wasn't real it is hard to know the actual stresses for that area. Perhaps a different ansys setup would have eliminated this stress and been more accurate.

For the bend test 0.3452 mm deformation was read when the bend test was conducted when it was believed that zero plastic deformation happened. Group six believe that this could be because the part used to apply the 50 N for the bending test was 3D printed and therefore could have experienced deformation itself and introduced some error into the final deformation read from the test.

Group Six believes that the failure of the wrench may have had to do with the clearance between the wrench and the bolt. This clearance allowed localized contact and for slippage which led to eventual stripping of the inside of the wrench. This clearance was not accounted for in the ansys analysis.

Section 08: Appendix A: Detailed Drawings



Section 09: Appendix B: Manufacturing Plan Summary

<i>Manufacturing Step</i>	<i>Step Description</i>	<i>Responsible Group Member</i>
<i>01</i>	Mark up aluminum stock part using a ruler and sharpie marker	<i>Zack</i>
<i>02</i>	Bring down the length to 6.75" inch with bandsaw	<i>Ari</i>
<i>03</i>	Bring down the length to 6.55" using the milling machine	<i>Ben</i>
<i>04</i>	Bring down the total thickness of the part to 0.21" using the milling machine; first on one side then flip to other side to remove thickness for an even surface finish	<i>Isaac</i>
<i>05</i>	Brush periodically and use the blow gun to clear metal chips, preventing defects (applicable to the whole procedure)	<u><i>Shared Responsibility</i></u>

06	Reduce the width down to 1” from starting value of 1.25” again with an even surface finish on both edges	<i>Zack</i>
07	Conduct the edge finding to prep for the interior I-Beam cuts/pockets	<i>Ari</i>
08	Cut the I-Beam pocket on the first side	<i>Ben</i>
09	Repeat the process for the second pocket cut on the opposite side	<i>Isaac</i>
10	Cut wrench hole, using 3 passes in the Z direction considering max allowed depth of cut	<i>Zack</i>
11	Measure 45 degree angle with angle finder and the part to cut the extra purpose tool, use small incremental milling passes	<i>Ari</i>
12	Create chamfers on the wrench head, use angle finder calipers and sharpie to mark up	<i>Ben</i>

13	Deburring of final wrench to remove sharp jagged edges	<i>Isaac</i>
14	Take weight: the final weight was recorded to be 1.15 ounces exactly in the machine shop	<u><i>Shared Responsibility</i></u>

Section 10: Appendix C: Summary of ANSYS FEA Results

Iteration 1:

Results:

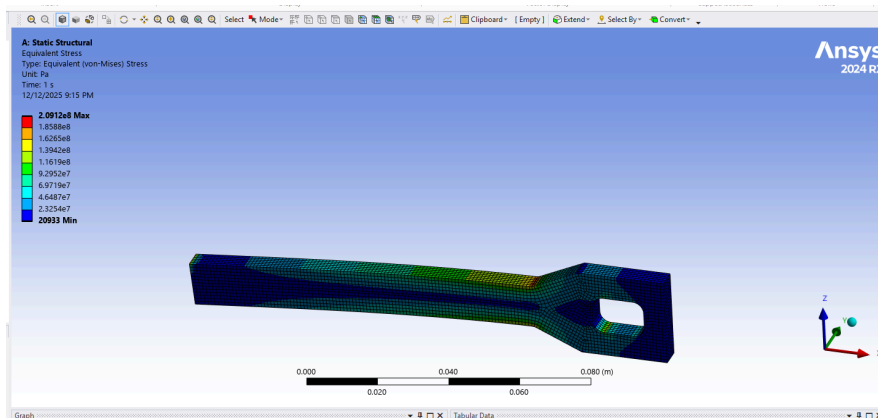
Mass = 36.35g

Torque = 190 Mpa

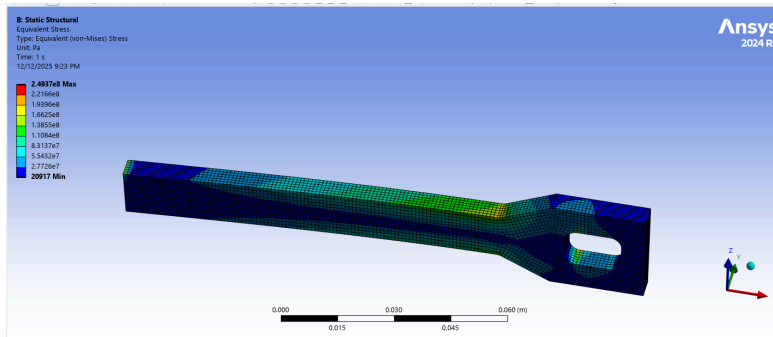
Combined = 197Mpa

Bending = 16 Mpa

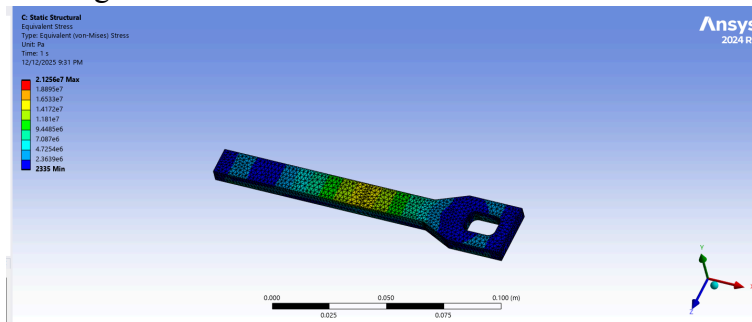
Torque Test:



Combined Test:



Bending Test:



Iteration 2:

Results:

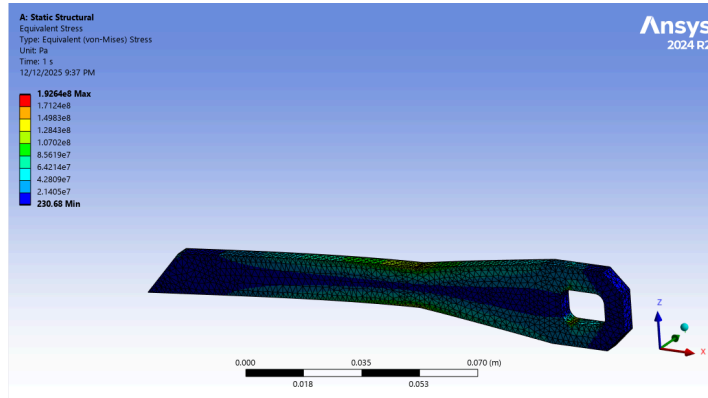
Mass = 32.54g

Torque = 145 Mpa

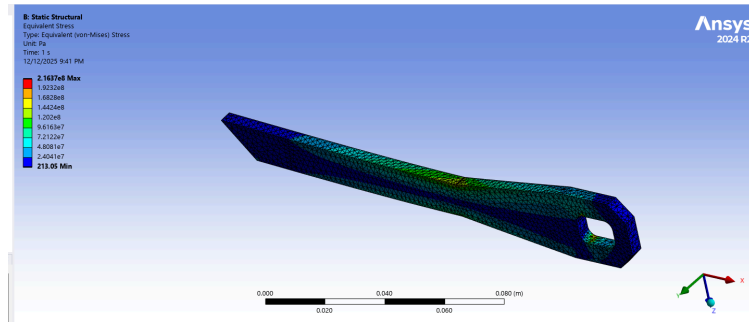
Combined = 155 Mpa

Bending = 22.5 Mpa

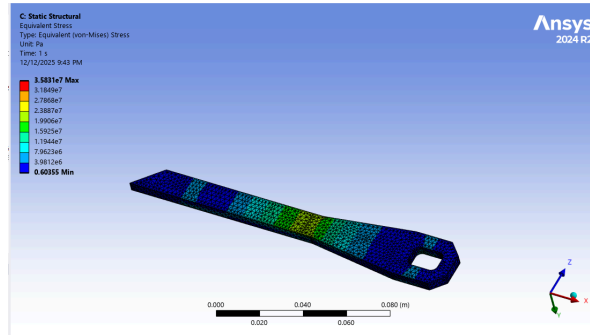
Torque Test:



Combined Test:



Bending Test:



Iteration 3:

Results:

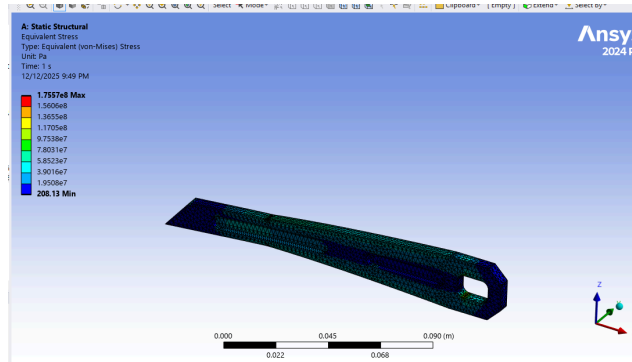
Mass = 31.98g

Torque = 72 Mpa

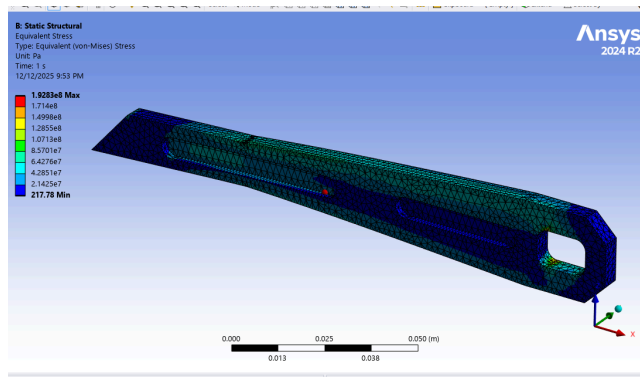
Combined = 81 Mpa

Bending = 15 Mpa

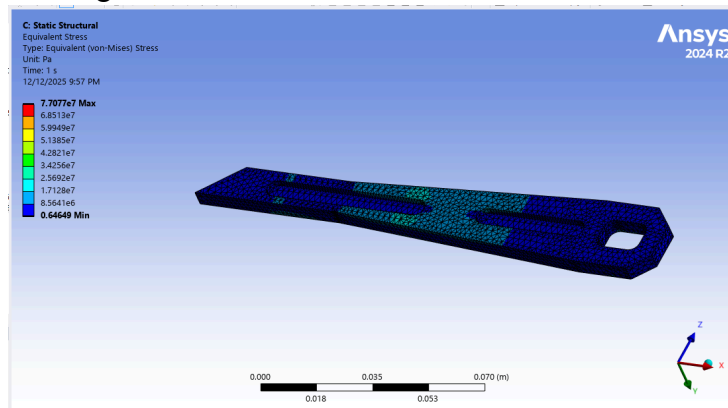
Torque Test:



Combined Test:



Bending Test:



Iteration 4:

Results:

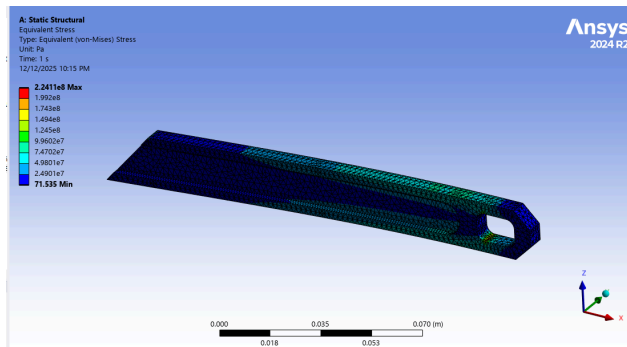
Mass = 29g

Torque = 81 Mpa

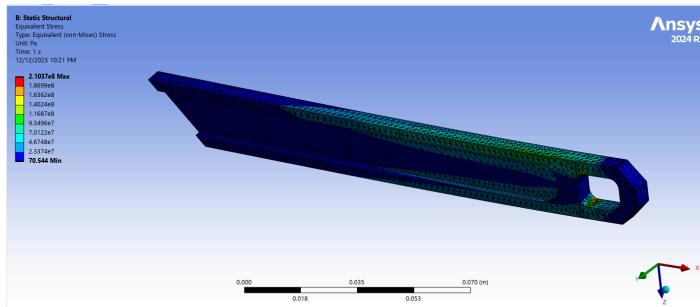
Combined = 100 Mpa

Bending = 50 Mpa

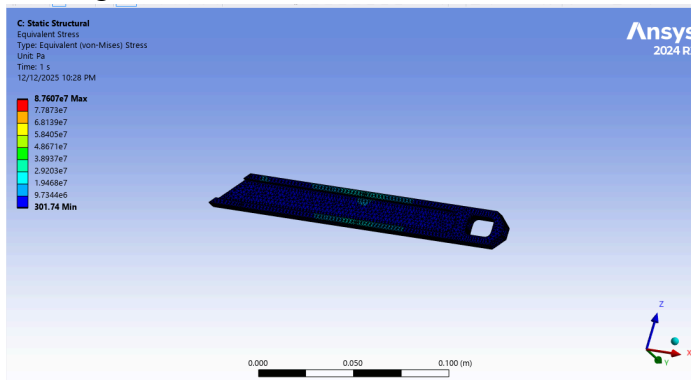
Torque Test:



Combined Test:



Bending Test



Iteration 5:

Results:

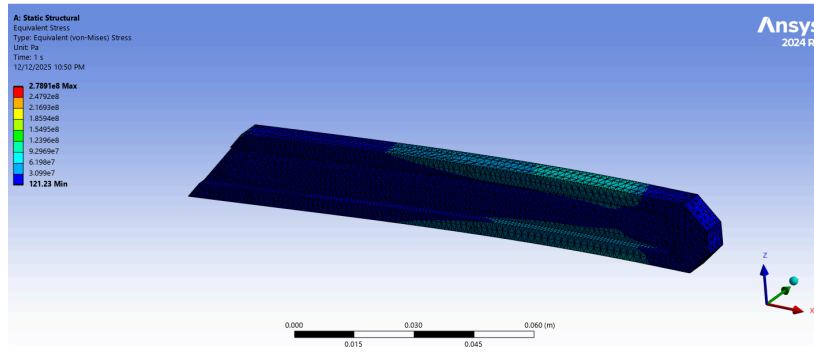
Mass = 30.77g

Torque = 85 Mpa

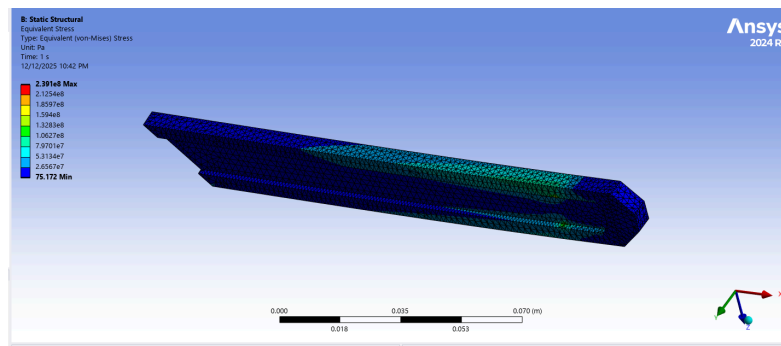
Combined = 96 Mpa

Bending = 13.8 Mpa

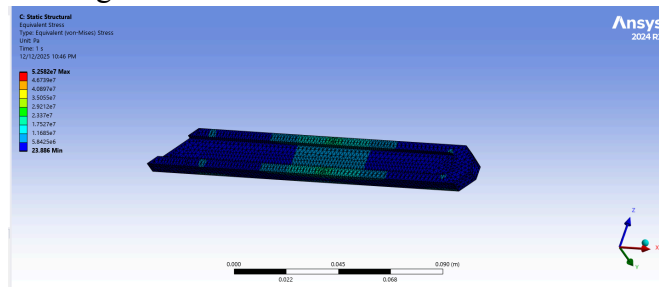
Torque Test



Combined Test



Bending Test



Iteration 6 (Final):

Results:

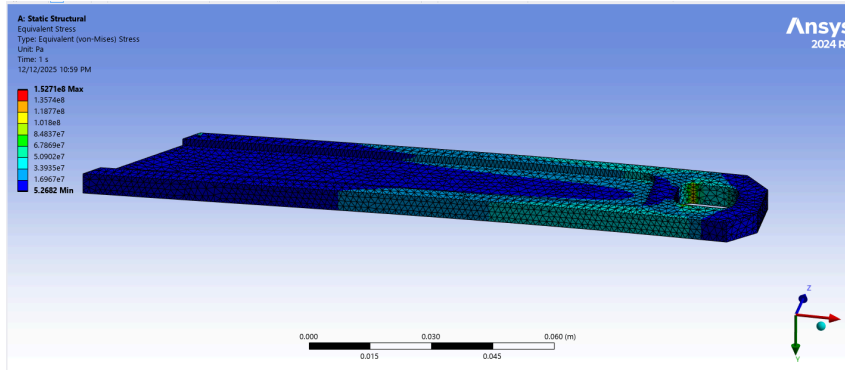
Mass = 31.9g

Torque = 45 Mpa

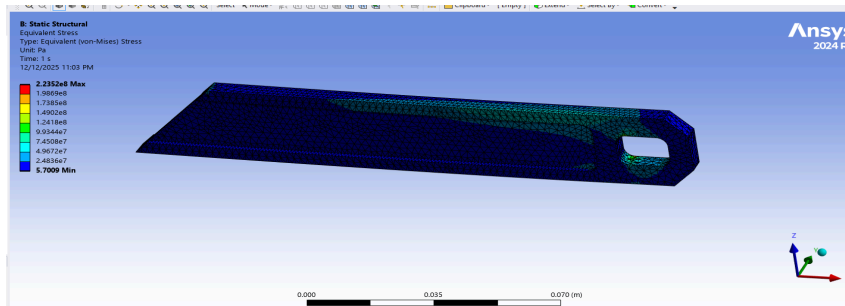
Combined = 73 Mpa

Bending = 26 Mpa

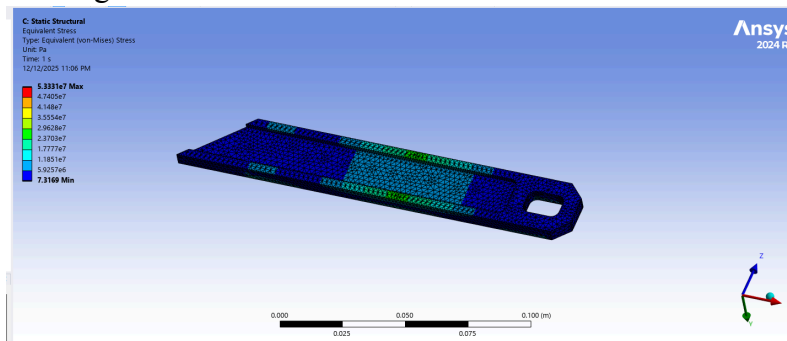
Torque Test



Combined Test



Bending Test



Section 11: Appendix D: Final Design

